Fitting ω Centauri with a compact action-based distribution function

Model, data issues, fitting performance and the status of the dark-matter question

oCen_dm project notes

23 September 2026

Abstract

We fit the surface-brightness profile and 89 binned velocity-dispersion measurements of ω Centauri (HST and Gaia EDR3 proper motions, MUSE line-of-sight velocities) with a self-consistent spherical equilibrium. Its stars follow a positive, regularized exponential action distribution function (DF); stellar remnants and an optional cored dark-matter (DM) halo enter as prescribed densities. The original DF, with a single orbital-anisotropy transition, cannot fit the combined data: the best fit has $\chi^2_{\rm kin}/N = 4.80$. All valid starts converge to the same branch minima, so the optimizer is not the cause. Wider search bounds, free distance and the HST/Gaia boundary are also excluded. The model fits HST+MUSE alone ($\chi^2_{\rm kin}/N = 1.43$) and Gaia alone (0.88), but these subsets demand opposite anisotropy: radial at small actions, tangential at large actions. A second anisotropy transition, already part of the model definition, removes most of the conflict. Without a halo it reaches $\chi^2_{\rm kin} = 159.2$ for 89 bins ($\chi^2_{\rm kin}/N = 1.79$), better than the best existing Jeans fit to the same bins ($\chi^2 = 190.0$). Its intrinsic anisotropy reproduces the independently inferred Jeans turnover out to ~ 30 pc. Remaining misfits are localized: the outer edge of the MUSE field and the tangential proper motions at 4–20 pc. Adding a free cored halo lowers the objective by 31 but drives two coordinates to their bounds and rearranges the anisotropy. This is a mass–anisotropy trade, not a DM measurement. We document the data issues that limit a statistical verdict: very small formal errors, approximate rotation corrections, a heterogeneous photometric splice and correlated bins. We also document the numerical defects found and fixed, and the programme needed to turn a good fit into a meaningful DM constraint.

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1 Aims

The project has two objectives, reported separately throughout:

1. **Fit the data well**: find a physically admissible equilibrium model that reproduces the light profile and all kinematic profiles within their uncertainties.
2. **Place meaningful constraints on the DM content**, chiefly the DM density at 20 pc, ρ_{20}.

The two objectives pull against each other. A more flexible stellar DF fits better, but it can also mimic a halo, which weakens the DM constraint. A model must pass objective 1 before any DM statement from it is meaningful. It should also be no more flexible than the data require.

This note covers the observed-data fits of 23 September 2026. The model definition is in `docs/compact_df_model.tex`, the implementation tests in `docs/REGULARIZED_DF_IMPLEMENTATION.md`, and the full log in `JOURNAL.md`.

2 Data

2.1 Datasets

All fits use one frozen data snapshot (fingerprint `37dfc7fa...`, file `results/df/pilot_no_dm_20260922/`). The same snapshot was used by the earlier Jeans fits, so their χ^2 values are directly comparable. Radii are converted at 0.02633 pc per arcsec, i.e. $D = 5.43$ kpc (the project's adopted distance, from Baumgardt and Vasiliev, 2021).

The kinematic likelihood uses the published asymmetric uncertainties in a split-normal χ^2. For bin i, with model m_i, datum d_i and errors $(\sigma_{i,-}, \sigma_{i,+})$,

$$\chi^2_{\rm kin} = \sum_i \left(\frac{m_i - d_i}{\sigma_{i,\pm}} \right)^2, \qquad \sigma_{i,\pm} = \begin{cases} \sigma_{i,+}, & m_i > d_i \\ \sigma_{i,-}, & m_i \le d_i \end{cases}. \tag{1}$$

Model predictions are averaged over each bin's saved radial selection nodes, in the units of the data. The photometric term uses a single profiled magnitude zero point (constant stellar mass-to-light ratio) and the adopted magnitude errors. The fitted objective is $\chi^2_{\rm kin} + \chi^2_{\rm phot}$.

Table 1: The observational inputs. R gives the range of bin centres, with bin edges in parentheses. "Rel. err" is the range of the published dispersion uncertainty relative to the dispersion. $\bar{v}^2/\sigma^2$ is the largest ratio of the adopted mean-streaming term to the squared dispersion (Section 2.2).

Dataset	n	R [pc]	Rel. err	max $\bar{v}^2/\sigma^2$	Provenance
HST PM radial	21	0.10–9.0 (0.09–9.5)	0.2–7.1%	0	Our measurement, `selection_hq_astrometry`, cluster+field mixture, to 360 arcsec
HST PM tangential	21	0.10–9.0 (0.09–9.5)	0.3–7.1%	0.25	As above; the Vasiliev and Baumgardt (2021) rotation curve is added to the model
MUSE LOS	29	0.15–7.6 (0–8.7)	1.4–11%	0.14	oMEGACat VI adaptive bins (Häberle et al., 2025), dispersion about the fitted rotation curve
Gaia EDR3 PM radial	9	9.4–56.6 (7.9–63.2)	1.4–10%	0	Our measurement, one component, raw errors
Gaia EDR3 PM tangential	9	9.4–56.6 (7.9–63.2)	1.9–10%	0.38	As above, published rotation
Surface brightness	82	0.045–68.1	0.10–0.58 mag	–	oMEGACat star counts ($F625W < 19$, 129,074 stars) inside 25 arcsec, Trager et al. (1995) V band outside

2.2 Data issues

The following properties of the data limit how the goodness of fit can be interpreted. None of them was altered in the fits reported here: all fits use the published errors (Section 7).

Very small formal kinematic errors. The HST PM dispersions reach 0.2–0.3% precision (e.g. $0.592 \pm 0.001\,\mathrm{mas\,yr^{-1}}$ at 7.1 pc). The Gaia dispersions have 1–2% errors (0.005–$0.007\,\mathrm{mas\,yr^{-1}}$) at 17–35 pc. The Gaia errors are purely statistical. Gaia EDR3 has spatially correlated astrometric systematics on small angular scales and underestimated per-star uncertainties in crowded fields (Lindegren et al., 2021; Vasiliev and Baumgardt, 2021). If per-star errors ϵ are underestimated, the deconvolved dispersion $\sigma^2_{\rm true} = \sigma^2_{\rm obs} - \epsilon^2$ is biased high. The effect is a bias in the measurement, not only in its error bar. At this precision, small calibration effects can dominate χ^2. We have not re-derived either profile with a calibrated error model. That is an open, separate data task.

Heterogeneous photometry. The photometric profile splices oMEGACat star counts (inside 25 arcsec, zero point offset by +16.991 mag over 30–100 arcsec) with the Trager et al. (1995) compilation. Its errors are adopted weights, $0.1\,\mathrm{mag}/\sqrt{w}$, not measured uncertainties; the splice covariance is omitted. Ten pairs of full-weight (0.1 mag) points at the same radius differ by up to 0.40 mag, i.e. 0.13 mag per point. A perfect model would therefore give $\chi^2_{\rm phot}/N \simeq 1.7$ on full-weight points (an estimate from ten pairs). Section 5.5 shows that the photometric misfit comes mainly from points of opposite sign at the same radii at 10–40 pc.

Rotation is subtracted, not modelled. The DF is non-rotating (even in L_z). The published dispersions are measured about a mean motion, so each model second moment is corrected as $\sigma_{\rm model} = \sqrt{\langle v^2\rangle - \bar{v}^2}$. For MUSE this uses $\bar{v}^2 = v^2_{\rm rot}/2$ from the fitted rotation curve; for the tangential PMs it uses the Vasiliev and Baumgardt (2021) curve. Adding an odd-L_z part to an even DF leaves the second moments unchanged. The subtraction is therefore exact for that model class, provided the measured dispersion is taken about the correct local mean. The approximations are geometric: the annular average, inclination, and sphericity. MUSE also reveals a counter-rotating

core (Pechetti et al., 2024). The correction is largest exactly where the fits are worst: $\bar{v}^2/\sigma^2$ reaches 0.38 in the Gaia tangential bins and 0.14 at the MUSE outer edge.

Correlated bins. The HST radial and tangential profiles are built from the same stars, as are the two Gaia components. Treating bins as independent makes absolute χ^2/N values and p-values indicative only.

The HST/Gaia boundary. The HST bins end at 9.5 pc and the Gaia bins begin at 7.9 pc (bin edges). In the one near-coincident pair of bins (9.0 and 9.4 pc) the surveys agree within errors: $\sigma_T = 0.465 \pm 0.027$ vs $0.474 \pm 0.049\,\mathrm{mas\,yr^{-1}}$ and $\sigma_R = 0.531 \pm 0.029$ vs $0.510 \pm 0.052\,\mathrm{mas\,yr^{-1}}$. Both errors are large, so this is only a weak consistency check.

Fixed distance. $D = 5.43$ kpc throughout, except in one diagnostic fit (Section 5.2).

3 Model

3.1 Stellar distribution function

Following `compact_df_model.tex` (Section 2), the stars follow a positive DF of the radial action J_r and total angular momentum $L = J_z + |J_\phi|$ in the total potential. Constant-DF contours are obtained by integrating

$$\frac{\mathrm{d}L}{\mathrm{d}J_r} = -g(J_r, L), \qquad g = g_H(J_r, L) \exp\left[-b(q) \sin\left(\frac{\pi c}{2}\right)\right], \qquad q = J_r + L, \quad c = \frac{L}{q}, \tag{2}$$

to the L axis, which defines a label $L_c(J_r, L)$. The DF is

$$f_\star(J_r, L) = A_\star \exp\left[-\left(\frac{L_c(J_r, L)}{J_0}\right)^\alpha\right], \qquad A_\star = \frac{M_\star}{(2\pi)^3 \mathcal{I}_\star}, \tag{3}$$

where $\mathcal{I}_\star$ is the numerically converged action-space normalization. The reference $g_H = \Omega_r/\Omega_t$ uses the exact frequencies of the current potential, making the $b = 0$ contours energy surfaces. The sine factor enforces the regularity condition $g \to 2$ as $L \to 0$ at any anisotropy amplitude (Binney, 2026). $b > 0$ favours radial orbits and $b < 0$ tangential orbits. $b(q)$ is an orbital-weight prescription, not $\beta(r)$.

The **one-transition** model (the original baseline) has

$$b(q) = b_{\mathrm{out}}\, S(q; J_a), \qquad S(q; J_t) = \frac{q^2}{q^2 + J_t^2}, \tag{4}$$

so $b \to 0$ for $q \ll J_a$ and $b \to b_{\mathrm{out}}$ for $q \gg J_a$. The **two-transition** model is

$$b(q) = b_{\mathrm{out}}\, S(q; J_a) + (b_{\mathrm{outer}} - b_{\mathrm{out}})\, S(q; J_{\mathrm{outer}}), \qquad J_{\mathrm{outer}} > J_a, \tag{5}$$

with limits 0, b_{out} and b_{outer}. Both forms were defined in the model proposal; eq. (5) was implemented and unit-tested before this work. Stars carry both mass and light (mass follows light). There is one stellar population.

3.2 What the DF family can represent

Each parameter set defines one spherical, self-gravitating equilibrium. From it the model predicts, with no further assumptions:

- the intrinsic stellar density $\rho_\star(r)$ and its projection $\Sigma(R)$, compared with the photometry through one zero point (constant M/L);

- the intrinsic radial and tangential dispersions $\sigma_r(r)$ and $\sigma_t(r)$ (one component), hence $\beta(r) = 1 - \sigma_t^2/\sigma_r^2$;
- the projected LOS dispersion $\sigma_{\rm los}(R)$ and the two plane-of-sky dispersions $\sigma_{{\rm pm},R}(R)$ and $\sigma_{{\rm pm},T}(R)$. Their ratio $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$ is the most direct projected signature of anisotropy;
- the enclosed mass of every component and the full phase-space density $f(J_r, L)$.

It predicts no mean rotation (the DF is even in L_z), no flattening and no population differences.

Figure 1 shows what shapes the family can take. Starting from the two-transition best fit (Section 5.3), one ingredient is varied per row and each equilibrium is rebuilt self-consistently in the fitted remnant potential:

- **Envelope exponent** α (row a) sets how steeply the DF, and hence the density, falls with action. At fixed J_0 and $M_\star$, moving from $\alpha = 0.7$ to 1.4 shrinks the half-mass radius from 22 to 2 pc and raises the central LOS dispersion from 15 to 40 $\mathrm{km\,s^{-1}}$. It also changes the anisotropy the same $b(q)$ produces, because the orbits populate a different range of q.
- **Action scale** J_0 (row b) mainly sets the size. The half-mass radius is 5, 10 and 24 pc for $J_0 = 40$, 65 and 120 $\mathrm{pc\,km\,s^{-1}}$. At fixed mass, the central dispersion falls as the cluster grows.
- **One anisotropy transition** (row c) gives only monotonic profiles. $b_{\rm out} > 0$ makes β rise towards radial outside J_a, so $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$ falls below one. $b_{\rm out} < 0$ gives tangential outskirts, with the ratio rising above one. A single transition cannot turn from radial to tangential. The anisotropy is not independent of the light: at fixed J_0 and α the half-mass radius changes from 6.7 to 18 pc between $b_{\rm out} = -1$ and $+1$. The photometry and kinematics therefore have to be fitted jointly.
- **The second transition** (row d) adds the non-monotonic shape the data require (Section 5.2): a mild radial bump followed by tangential outskirts, seen in projection as a dip and then a rise of $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$. $J_{\rm outer}$ sets where the turnover occurs. At the best fit ($J_{\rm outer} = 812$ $\mathrm{pc\,km\,s^{-1}}$), β peaks at $+0.16$ near 9 pc and crosses zero at 23 pc. With $J_{\rm outer} = 300$ $\mathrm{pc\,km\,s^{-1}}$ the peak is only $+0.04$ (at 2 pc) and the zero crossing moves in to 4 pc.

A built-in property: every model is isotropic at the centre. $b(q) \to 0$ as $q \to 0$, and the regularity condition forces $g \to g_H$ there, so $\beta(0) = 0$ for every parameter choice. The family cannot produce a tangentially or radially biased core. Other built-in properties are positivity, smoothness in action space, and finite total mass. The central limitation matters when comparing with Jeans models, whose central anisotropy is free: the rung-2 Jeans fit has $\beta_0 = -0.5$, set by its prior.

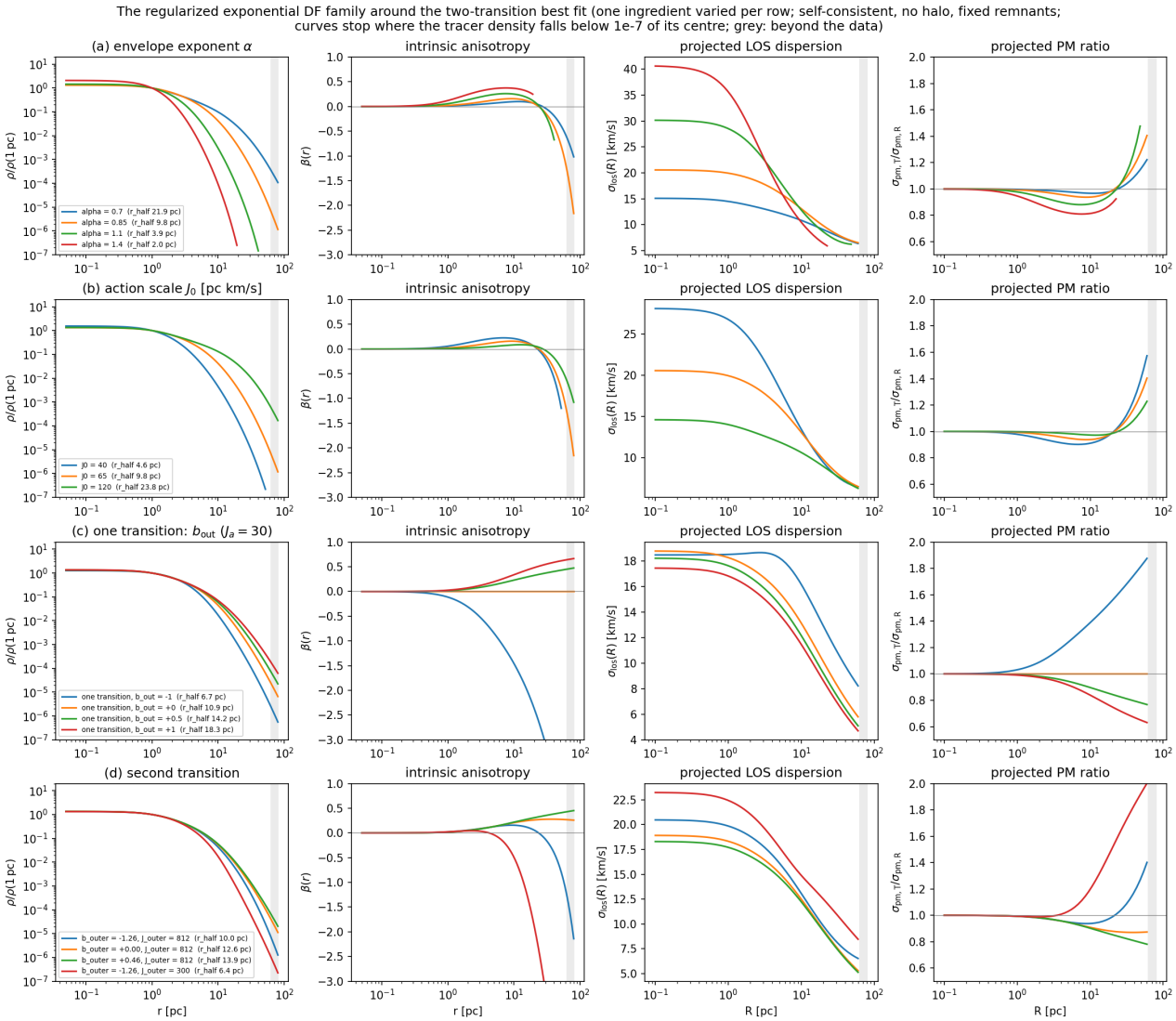


Figure 1: The regularized exponential DF family around the two-transition best fit. Rows vary (a) α, (b) J_0, (c) a single anisotropy transition with amplitude $b_{\rm out}$ at $J_a = 30\,\mathrm{pc\,km\,s^{-1}}$, and (d) the second transition ($b_{\rm outer}$, $J_{\rm outer}$). All other parameters stay at the best-fit values. Columns: intrinsic density normalized at 1 pc (legend: half-mass radius); intrinsic $\beta(r)$; projected $\sigma_{\rm los}(R)$; projected $\sigma_{{\rm pm},T}/\sigma_{{\rm pm},R}$. Every model is a self-consistent equilibrium in the fitted remnant potential (no halo, no point mass). Curves stop where the tracer density falls below 10^{-7} of its central value. Grey marks radii beyond the data. These are illustrations, not fits (script `bin/plot_df_family_gallery.py`).

3.3 Dark components

Dark mass enters as prescribed spherical densities in the same potential:

- **Remnants**: a Plummer sphere with mass $M_{\rm rem}$ and scale $a_{\rm rem}$.
- **Halo** (free-halo branch only): a cored profile ($\gamma = 0$) normalized at 20 pc, $\rho_{\rm DM}(r) = \rho_{20}\, h(r/r_s)T(r/r_t)/[h(20\,{\rm pc}/r_s)T(20\,{\rm pc}/r_t)]$, with $h(x) = (1+x)^{-3}$ and $T(y) = e^{-y^2}$. The scale r_s is free and the taper $r_t = 1000$ pc is fixed. The no-halo branch sets $\rho_{20} = 0$.
- **No central point mass** (Section 5.6).

3.4 Equilibrium, projection and numerics

For every trial, the stellar DF is rebuilt and iterated with Poisson's equation, with the dark densities held fixed, until the stellar density closes (AGAMA: Vasiliev, 2019). Projected surface density and LOS/PM second moments follow from the intrinsic moments, averaged over each bin's selection nodes. The fitting resolution uses 128 potential nodes, 32 velocity nodes, 240 moment nodes, 128 projection nodes and 192 action-contour nodes. The equilibrium iteration tolerance is 5×10^{-5} (Section 4.2). Every selected solution is validated at doubled resolution.

4 Fitting method and validation

4.1 Optimizer

Fits use bounded trust-region least squares (SciPy `least_squares`, TRF) on the residual vector of eq. (1) plus photometry. All coordinates are scaled to $[0, 1]$; mass and action scales are logarithmic. Jacobians are forward differences with step 0.002, each probe seeded with the base point's converged stellar potential. The 9–11 probes of a Jacobian run in parallel worker processes. A fit stops when the objective improves by less than 0.01 over two accepted steps (well below a meaningful $\Delta\chi^2 \sim 1$), when SciPy's own tests are met, or at 600 equilibrium evaluations. Stops that follow closely after rejected equilibria are recorded as unconverged. Several starting points are used for each model: two hand-picked and, for the wide-bound and two-transition runs, four Latin-hypercube starts (seed 20260923) in the central 60% of each range.

The second transition is parametrized by $\Delta = \ln(J_{\rm outer}/J_a) \in [\ln 2, \ln 100]$, applied after all other coordinates, so $J_{\rm outer} > J_a$ holds for every trial. The search bounds are listed in Table 2. They are optimizer limits, not priors.

Table 2: Search bounds. The "default" bounds were used in the first batch only.

Coordinate	Bounds	Notes
$M_\star$	10^6–$6 \times 10^6\,{\rm M}_\odot$ (log)	
J_0	30–800 pc km s^{-1} (log)	default 100–800
α	0.6–3	
$b_{\rm out}$	-2 to $+2$	± 1 in one-transition runs
J_a	5–500 pc km s^{-1} (log)	default 50–3000; 5–3000 in wide one-transition runs
$b_{\rm outer}$	-4 to $+2$	two-transition only
$\ln(J_{\rm outer}/J_a)$	$\ln 2$–$\ln 100$	two-transition only
$M_{\rm rem}$, $a_{\rm rem}$	10^2–$10^6\,{\rm M}_\odot$, 0.3–20 pc (log)	
ρ_{20}, r_s	0–10 ${\rm M}_\odot$ pc^{-3}, 5–500 pc (log)	free-halo only; default $r_s \le 150$

4.2 Numerical checks

- **Evaluation noise.** At a converged model, repeating the equilibrium with different seeds shifted residuals by up to 0.0060 errors at the original tolerance (5×10^{-4}), and Jacobian columns by 0.1–7%. At 5×10^{-5} these fall to 0.00059 errors and 0.01–0.65%, at 23% higher cost per evaluation. The tighter tolerance is used throughout.
- **Parallel Jacobians** reproduce serial ones: all probe objectives agree to 2.4×10^{-9} relative. The potential is transferred by file, a round trip exact to 3×10^{-14}. Probe wall time drops 4.6-fold.
- **Validation of each selected fit**: a cold replay, a rebuild at doubled resolution, and an independent direct projection over the data radii (0.05–68 pc). The thresholds are shifts below 0.1 errors and projection agreement within 0.5%. All solutions reported here pass. The largest values are a cold shift of 0.0046, a refinement shift of 0.012 errors, and a projection difference of 0.11%.

4.3 Acceptance criteria

The formal errors are known to be incomplete (Section 2.2), so structural and statistical success are judged separately.

- **Structural pass:** independent starts converge; no coordinate at a search bound; no coherent residual pattern; no dataset with $\chi^2/n > 3$; a physically smooth DF.
- **Statistical pass:** $\chi^2_{\rm kin}/N < 1.3$, every dataset $\chi^2/n < 2$, $\chi^2_{\rm phot}/N < 2$. This is required only under a justified error model; with the published errors it is reported, not required.

5 Results

5.1 The one-transition DF fails

Table 3 summarizes the one-transition fits. In every branch, all valid starts converge to the same minimum; five of five do so in each wide-bound branch. The kinematic fit remains poor. The anisotropy scale J_a sits at its lower bound in every fit, whether that bound is 50 or 5 pc km s^{-1}. Widening the bounds changes $\chi^2_{\rm kin}$ by only 29 (free halo) and 20 (no halo). Figure 2 shows the residual pattern. The tangential PM dispersion is too high at 3–7 pc (to −6 errors) and too low beyond 12 pc (to +6). The LOS dispersion is 3–5 errors too high at 5–8 pc, and the radial PM 2–5 errors too high beyond 20 pc.

Table 3: One-transition fits. The objective is $\chi^2_{\rm kin} + \chi^2_{\rm phot}$ after refinement. Dataset columns give χ^2/n.

Fit	Obj.	$\chi^2_{\rm kin}$	$\chi^2_{\rm kin}/N$	HST R	HST T	MUSE	Gaia R	Gaia T	At bound
Default, free halo	653.6	456.1	5.12	2.5	5.6	2.7	8.2	15.1	J_0, J_a, r_s
Default, no halo	717.6	475.7	5.34	2.3	6.8	2.8	2.6	20.1	J_a
Wide, free halo	623.3	427.5	4.80	2.1	4.5	2.6	10.1	13.6	J_a
Wide, no halo	698.1	456.1	5.12	2.1	6.2	2.7	2.6	20.2	J_a

5.2 Why it fails: diagnostic fits

Diagnostic fits of the same no-halo DF (Table 4) isolate the cause.

1. **The DF fits each half of the data.** HST+MUSE and Gaia are each fitted well by the same family. Only their combination fails.

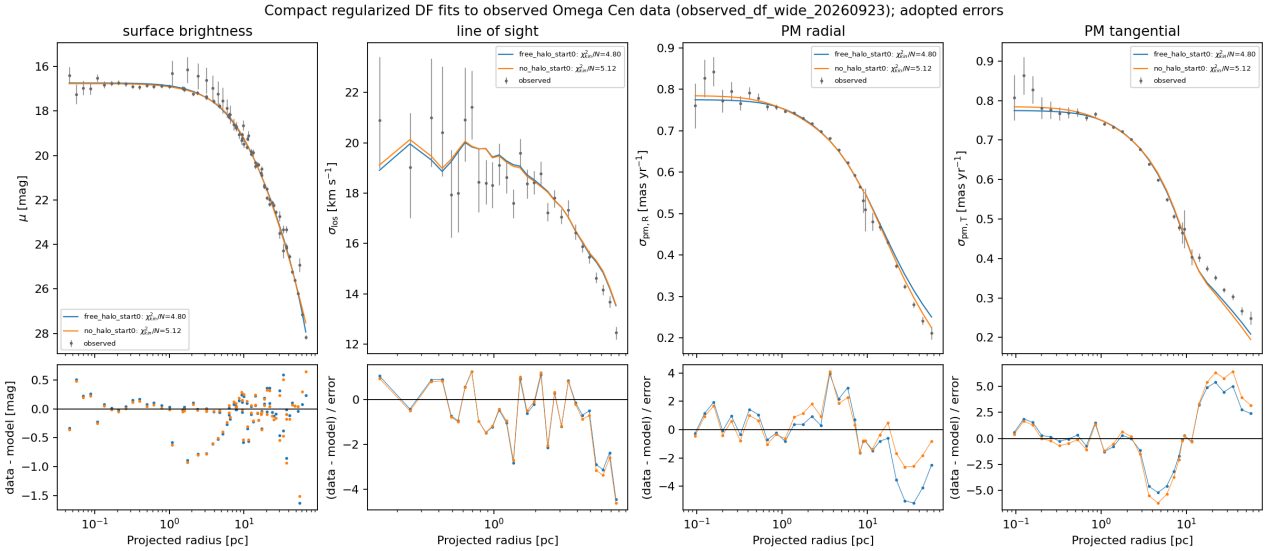


Figure 2: Best one-transition fits (wide bounds) with and without a halo. Top: surface brightness, LOS and PM dispersions with published errors. Bottom: normalized residuals (data−model)/error.

Table 4: Diagnostic fits of the one-transition, no-halo DF. Both starts of every fit converged to the same solution. The Jeans row is the rung-2 turnover fit (with a point mass, no DM) on the same 89 bins. (b): at a search bound.

Fit	$\chi^2_{\rm kin}/N$	HST R	HST T	MUSE	Gaia R	Gaia T	$b_{\rm out}$	J_a
All data	5.12	2.1	6.2	2.7	2.6	20.2	+0.18	5 (b)
HST+MUSE only (71 bins)	**1.43**	1.0	1.1	2.0	–	–	+0.39	32
Gaia only (18 bins)	**0.88**	–	–	–	0.8	1.0	−1.00 (b)	1665
All data, distance free	4.78	1.8	6.6	1.6	2.6	19.6	+0.18	5 (b)
Jeans turnover (ref.)	*2.13*	*2.9*	*1.7*	*1.8*	*1.9*	*2.6*		

2. **The halves demand opposite anisotropy.** The inner data want radial bias switching on at small actions ($b_{\rm out} = +0.39$, $J_a = 32\,{\rm pc\,km\,s^{-1}}$). The outer data want strongly tangential bias ($b_{\rm out} = -1$). One transition cannot supply both.

3. **Distance is not the cause.** Freeing D gives $D = 5.30$ kpc and lowers $\chi^2_{\rm kin}$ by only 31. Changing D rescales both PM components equally and cannot change σ_T/σ_R.

4. **An independent Jeans fit needs the same shape.** The rung-2 Jeans turnover fit prescribes $\beta(r) = 1 - \sigma_t^2/\sigma_r^2$ with two transitions. It finds $\beta \simeq +0.15$ near 6 pc, zero near 20 pc, and -0.6 to -0.7 at 80 pc. Its central value, $\beta_0 = -0.50$, sits at its prior limit (Figure 3).

The tangential turn lies at ~ 20 pc, inside the Gaia coverage, not at the HST/Gaia boundary.

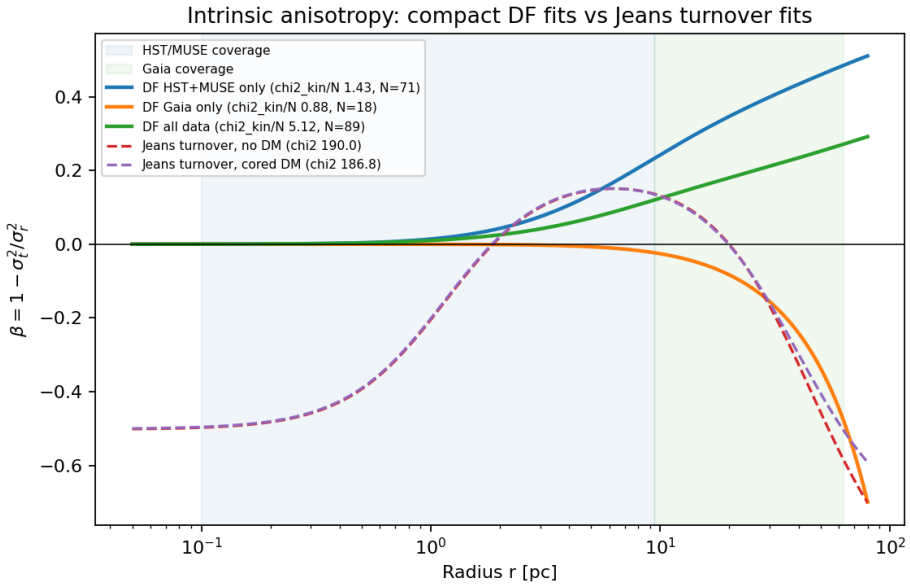


Figure 3: Intrinsic anisotropy $\beta(r)$ of the one-transition DF fitted to HST+MUSE only, to Gaia only, and to all data, compared with the rung-2 Jeans turnover fits. Shading marks the radial coverage of the two surveys.

Action coverage. The anisotropy acts on $q = J_r + L$. Sampling 2×10^5 tracers from the best all-data DF shows that the data sample $q \simeq 20$–$730\,{\rm pc\,km\,s^{-1}}$ (5–95%; Figure 4). The Gaia-only $J_a = 1665$ lies above even the outermost bin's 95th percentile. That fit therefore used only the low-q tail, $b \simeq b_{\rm out}(q/J_a)^2$, where only $b_{\rm out}/J_a^2$ is constrained. Neighbouring radial ranges overlap strongly in q, so a transition in q is soft in radius. These results set the bounds for the second transition (Table 2).

5.3 The two-transition DF without a halo

Six starts were run: the two hand-picked starts combine the two subset solutions, and four are Latin-hypercube starts. Two Latin starts were infeasible at their initial point. The four valid starts converged to one minimum (refined objective 378.3–378.4, $\Delta\chi^2_{\rm kin} \leq 0.6$), with no coordinate at a bound and no rejected evaluation. The best-fit parameters are $M_\star = 3.25 \times 10^6\,{\rm M_\odot}$, $J_0 = 65\,{\rm pc\,km\,s^{-1}}$, $\alpha = 0.85$, $b_{\rm out} = +0.46$, $J_a = 27\,{\rm pc\,km\,s^{-1}}$, $b_{\rm outer} = -1.26$, $J_{\rm outer} = 812\,{\rm pc\,km\,s^{-1}}$, $M_{\rm rem} = 1.72 \times 10^5\,{\rm M_\odot}$ and $a_{\rm rem} = 1.27$ pc.

$\chi^2_{\rm kin}$ falls from 456.1 to **159.2** ($\chi^2_{\rm kin}/N = 1.79$). Gaia tangential improves from $\chi^2/n = 20.2$ to 3.7 (Table 5). The intrinsic $\beta(r)$ follows the Jeans turnover to ~ 30 pc (Figure 5): $\beta = 0.02$, 0.14, 0.16, 0.06 and -0.13 at 1, 6, 10, 20 and 30 pc. Beyond ~ 40 pc it falls faster, reaching -1.27 at 63 pc. That region is driven by the $b_{\rm outer}$ asymptote, which acts only beyond the sampled q, so it is an extrapolation. The bias $b(q)$ peaks at $+0.41$ at $q = 103\,{\rm pc\,km\,s^{-1}}$, where the HST/MUSE tracers

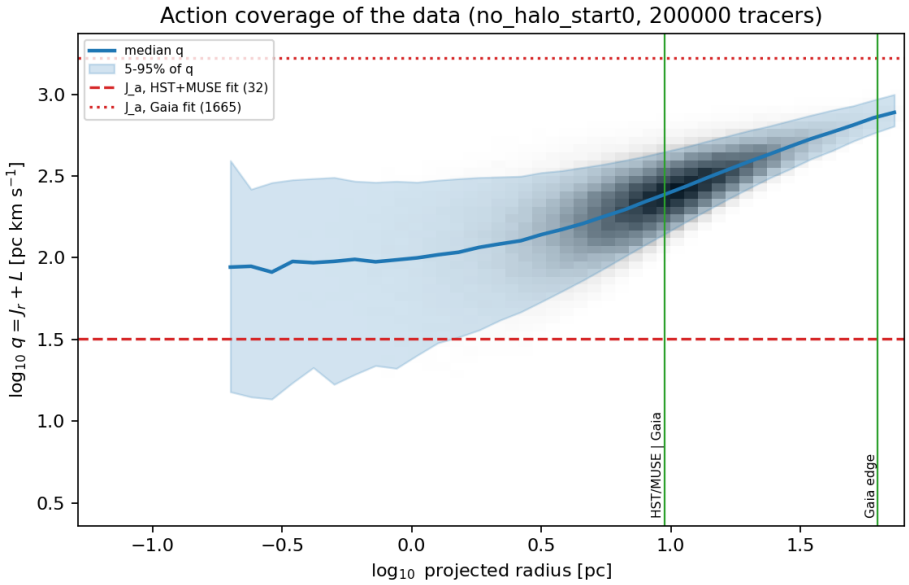


Figure 4: Distribution of $q = J_r + L$ of DF tracers against projected radius, with 5–95% quantiles. Horizontal lines mark the J_a of the HST+MUSE-only (32) and Gaia-only (1665) fits.

lie (q 5/50/95% = 56/170/379). It crosses zero at $q = 488$, inside the Gaia tracers (183/347/650; Figure 6). The DF is smooth and monotonic in the (J_r, L) plane, with no ridge at either transition. The remnant component returns to the inner-data values: the $6.6 \times 10^5\,\mathrm{M}_\odot$ remnant of the Gaia-only fit was compensating for the missing tangential outskirts.

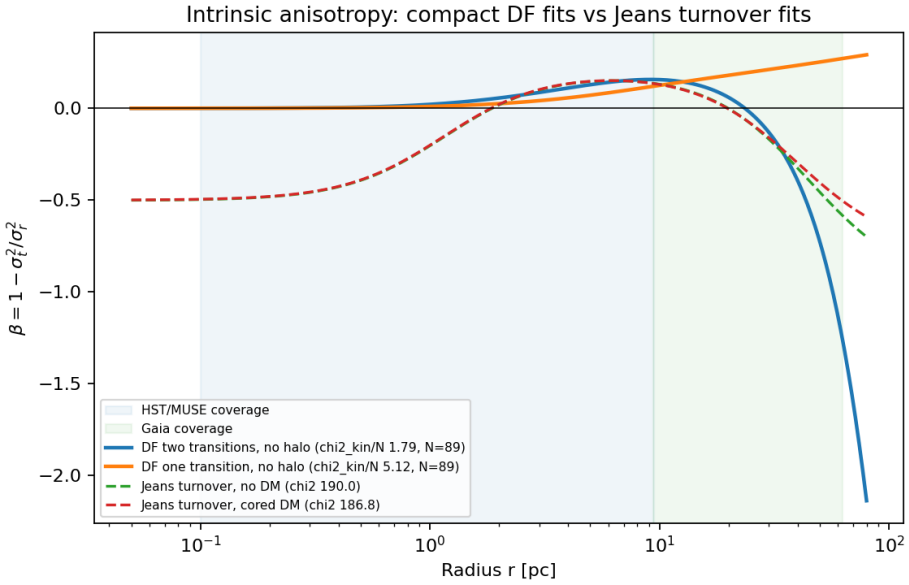


Figure 5: Intrinsic anisotropy of the two-transition fit (no halo) against the one-transition fit and the Jeans turnover fits.

5.4 Adding a free cored halo

With a free halo, three of six starts were infeasible at their initial point: two numerically (an action-normalization error of 5.1×10^{-5} above the 2×10^{-5} tolerance, and a density-closure failure) and one physically (the streaming term exceeded the DF second moment). The three valid starts converged to one minimum. The refined objective is 346.9, against 378.3 without a halo, and $\chi^2_{\mathrm{kin}} = 143.3$. The fitted parameters are $\rho_{20} = 0.92\,\mathrm{M}_\odot\,\mathrm{pc}^{-3}$, $M_\star = 3.21 \times 10^6\,\mathrm{M}_\odot$, $b_{\mathrm{out}} = 1.76$, $J_a = 101\,\mathrm{pc\,km\,s}^{-1}$, $b_{\mathrm{outer}} = -0.25$ and $J_{\mathrm{outer}} = 203\,\mathrm{pc\,km\,s}^{-1}$. However:

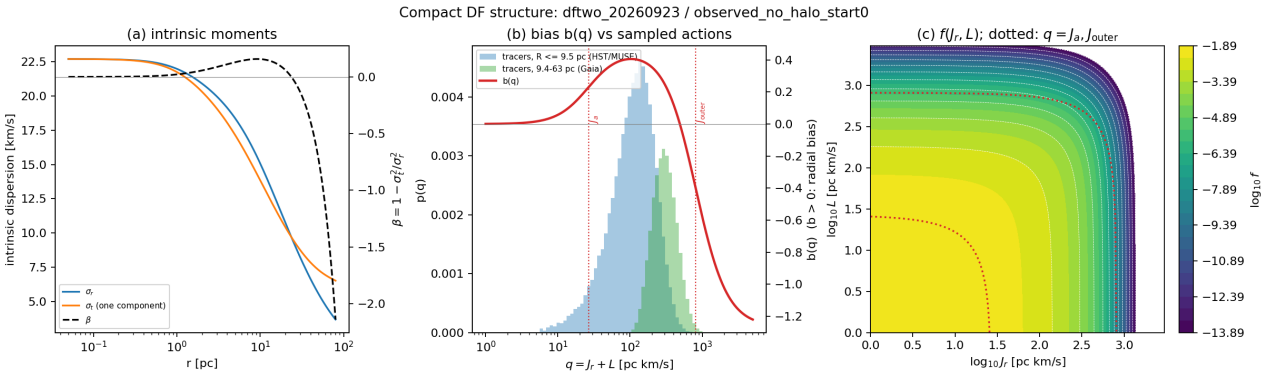


Figure 6: Structure of the two-transition fit (no halo). (a) Intrinsic σ_r, one-component σ_t and $\beta(r)$. (b) The bias $b(q)$ over the q distributions of tracers in the HST/MUSE and Gaia projected ranges. (c) $\log_{10} f$ in the (J_r, L) plane ($J_z = 0$); dotted lines mark $q = J_a$ and $q = J_{\rm outer}$.

- **Two coordinates are at bounds**: $r_s = 500\,\mathrm{pc}$ (upper), making the halo effectively uniform inside the data, and $J_{\rm outer}/J_a = 2$ (lower), pushing the two transitions together.
- **The anisotropy rearranges**: $b_{\rm out}$ changes from 0.46 to 1.76 and $b_{\rm outer}$ from -1.26 to -0.25.
- **The halo acts only in the outskirts**: it is under 1% of the enclosed mass at 20 pc, 8% at 42 pc and 32% at 80 pc. It trades misfit between the Gaia components: tangential improves from 3.7 to 2.3 while radial worsens from 0.9 to 2.0.

This is the classic mass–anisotropy trade. The objective gain (31, with $\Delta\chi^2_{\rm kin} = 16$ for two extra coordinates) cannot be read as a DM preference (Section 6).

5.5 Performance of the best-fit models

Table 5 and Figures 7–8 summarize the two best fits. The runs test counts sign changes of the residuals along radius; $z < -2$ means too few runs, i.e. coherent residuals.

Table 5: Performance of the two-transition fits with published errors. p is the χ^2 survival probability for n independent bins (indicative only). The last column gives the largest absolute normalized residual.

	no halo		free halo		runs z	max$\lvert r\rvert$
Dataset (n)	χ^2/n	p	χ^2/n	p	(no halo / halo)	(no halo / halo)
HST PM R (21)	1.01	0.45	1.02	0.43	-0.9 / 0.0	2.0 / 2.0
HST PM T (21)	1.81	0.013	1.39	0.11	-1.1 / -1.1	2.6 / 2.1
MUSE LOS (29)	2.01	0.001	1.84	0.004	+0.5 / +0.5	3.6 / 3.2
Gaia PM R (9)	0.93	0.50	2.04	0.03	-2.3 / -1.2	1.9 / 2.2
Gaia PM T (9)	3.70	10^{-4}	2.34	0.01	-0.8 / -1.0	3.6 / 2.7
Photometry (82)	2.67	2×10^{-14}	2.48	3×10^{-12}	-1.2 / -2.1	5.7 / 5.8
$\chi^2_{\rm kin}$ (89)		159.2		143.3		
$\chi^2_{\rm phot}$ (82)		219.1		203.6		

- **The kinematics are now fitted better than by the best Jeans model** ($\chi^2_{\rm kin} = 159.2$ and 143.3, against 190.0), with a positive, self-consistent DF. All profiles are tracked over 0.1–63 pc. The residual distribution is somewhat wider than $N(0,1)$, with tails to ±3.6.
- **The remaining structured misfits are localized.** The MUSE LOS is 2–3.6 errors too high at its outer edge (5–8 pc), where the streaming correction is largest. The tangential PMs are 2–2.6 errors too high at 4–7 pc and up to 3.5 errors too low at 15–20 pc (the first Gaia bins). With the halo, the outer Gaia radial PM becomes ~ 2 errors too high.

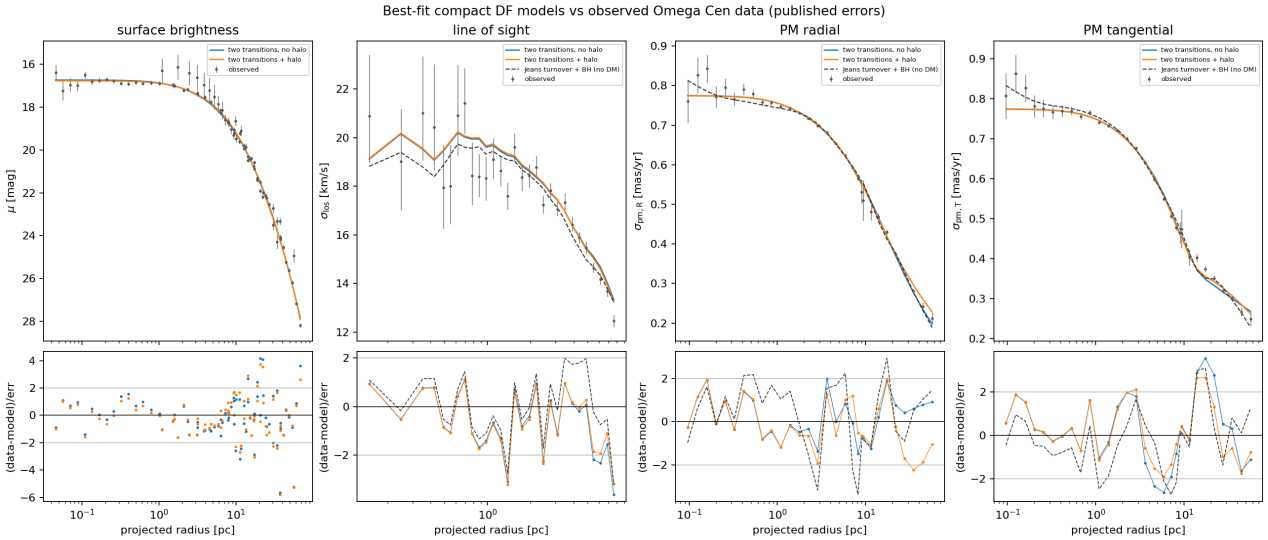


Figure 7: Best two-transition fits without (blue) and with (orange) a halo, and the rung-2 Jeans turnover fit (dashed; kinematics only). Bottom: (data−model)/error, with ±2 lines.

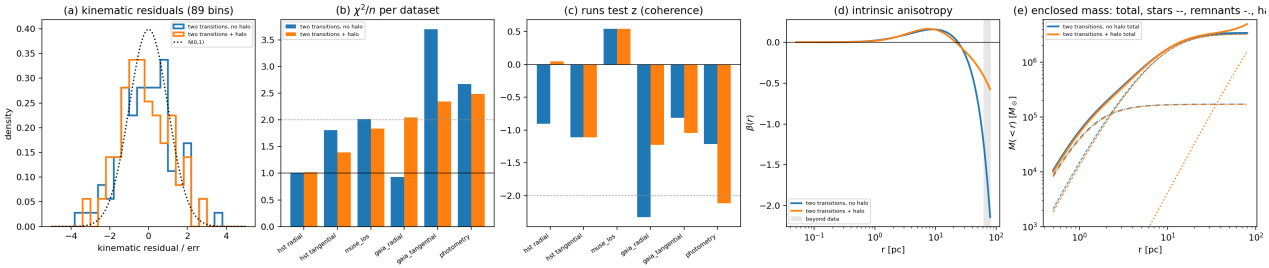


Figure 8: Diagnostics of the two best fits: (a) kinematic residuals against $N(0,1)$; (b) χ^2/n per dataset; (c) runs-test z per dataset; (d) intrinsic $\beta(r)$; (e) enclosed mass by component (total, stars, remnants, halo).

- **The photometric misfit** is dominated by 10–40 pc points with residuals of opposite sign at similar radii (−3 to −6 and +2 to +4). This is consistent with the internal scatter of the compilation (Section 2.2); this interpretation has not been tested.
- **The two models are nearly identical inside the data**: $\beta(r)$ and the total $M(<r)$ coincide to ~ 25 pc.

Verdict on objective 1. The two-transition DF is a structural near-pass. The only structural criterion it misses is Gaia tangential at $\chi^2/n = 3.7$ (without a halo), against a limit of 3. It is not a statistical pass under the published errors: $\chi^2_{\rm kin}/N = 1.79$ and 1.61.

5.6 Is a central point mass needed?

The model has no central black hole. Figure 9 compares the 43 kinematic bins inside 2 pc with the no-BH two-transition DF and the rung-2 Jeans fits, which include a free point mass with maximum-likelihood $M_\bullet = 4.4 \times 10^4\,\mathrm{M}_\odot$ (16–84%: 3.9–4.6×10^4).

Table 6: χ^2 of the 43 kinematic bins inside 2 pc.

Model	total	HST R (13)	HST T (13)	MUSE (17)
DF, two transitions, no BH, no halo	**48.8**	11.7	12.5	24.6
Jeans turnover + BH, no DM	59.0	21.5	16.7	20.8
Jeans turnover + BH, cored DM	59.3	23.7	14.1	21.4

The binned data give no evidence that the DF needs a point mass. Only two HST bins (0.12–0.16 pc) lie 1–2 errors above the no-BH model, and the innermost bin (0.10 pc) lies below it. The comparison is not controlled: the Jeans fits also differ in anisotropy (with $\beta_0 = -0.5$ imposed) and in mass model. The clean test would be the same DF with and without a BH. The binned profiles also cannot test the published IMBH evidence. That evidence rests on individual fast-moving stars inside ~ 0.1 pc, which set a lower limit of $\sim 8200\,\mathrm{M}_\odot$ (Häberle et al., 2024), and those stars are not in these profiles. Adding a BH is therefore deferred.

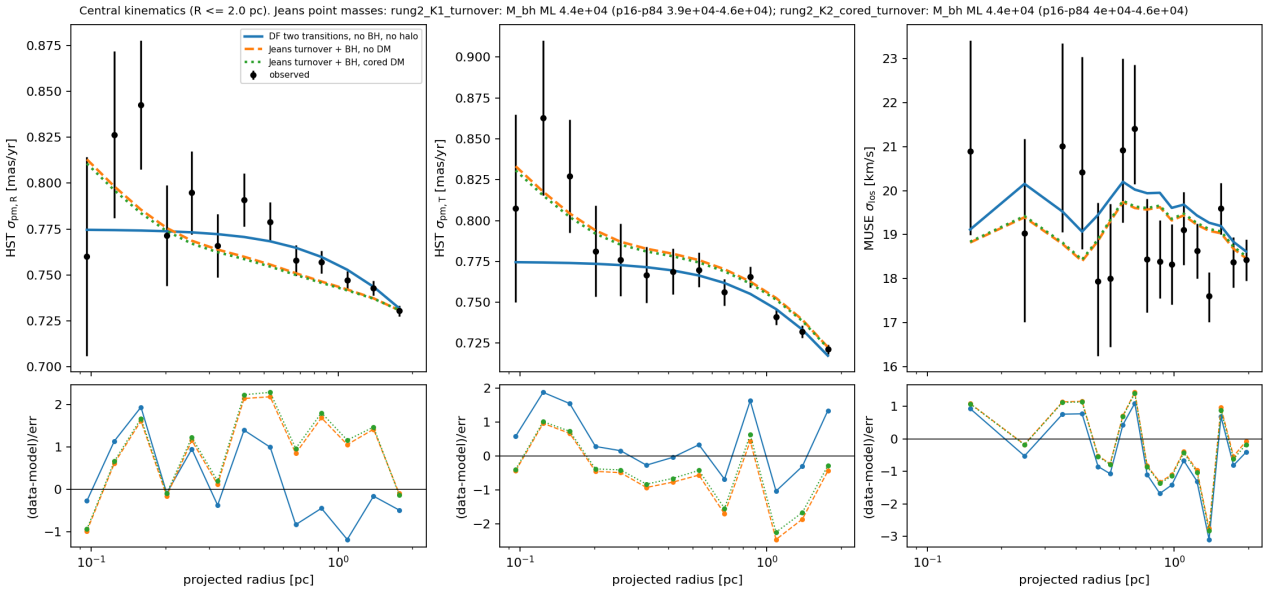


Figure 9: Central kinematics ($R \leq 2$ pc) with the no-BH two-transition DF and the Jeans turnover fits with a point mass. Bottom: (data−model)/error.

6 The dark-matter question: what can and cannot be said

No DM constraint yet. The halo improves the objective by 31. This cannot be read as a detection or as a preference. The halo solution has two coordinates at bounds; the anisotropy rearranges to accommodate it; the halo acts mainly beyond the outermost data; and the likelihood is not calibrated (Section 2.2).

The stellar/DM split is weakly identified. Noiseless mock tests with the one-transition family fitted a matched family (`docs/COMPACT_DF_RECOVERY.md`). The fits reproduced the observables to 0.006–0.02 errors RMS and recovered the total mass within 5%. They did not recover the split. In a no-DM mock the fits found $\rho_{20} = 1.4$–$2.3\,\mathrm{M_\odot\,pc^{-3}}$ with 15–24% less stellar mass; in a cored mock ($\rho_{20} = 2$) they found 2.5 and 3.1. The objective at the injected truth (2.3×10^{-4}) was lower than at the fitted solutions (6×10^{-3}–0.1), but by amounts far below the χ^2 scatter expected with real noise, $\sqrt{2 \times 171} \simeq 18.5$. Those mocks were also not ω Cen-like: their dispersions were 35–40% too low.

What a meaningful constraint requires.

1. A model that passes objective 1, with no coordinate at a bound.
2. A **profile likelihood in** ρ_{20}: fix ρ_{20} on a grid (e.g. 0, 0.25, 0.5, 1, 2, $4\,\mathrm{M_\odot\,pc^{-3}}$), refit all other coordinates including the anisotropy, and record the objective. r_s is a nuisance parameter that the data do not constrain; the quoted quantity should be ρ_{20} or $M_{\mathrm{DM}}(< r)$ within the data range.
3. **Validation on realistic mocks** built from the best no-halo and halo fits, with noise, to measure identifiability and coverage.
4. A **calibrated likelihood** before converting $\Delta\chi^2$ into intervals. Until the Gaia error model is settled, mock-calibrated thresholds are the substitute.

7 Numerical and software issues found and fixed

- **Optimizer budget and tolerances.** The first recovery batch capped fits at 180 evaluations, about 18 Gauss–Newton steps with nine coordinates. Its relative tolerances were tighter than the evaluation noise. An absolute stopping rule, a 600-evaluation cap and a tenfold tighter equilibrium tolerance fixed both problems.
- **Rejected equilibria in Jacobians.** A rejected probe previously entered as a 10^6 residual. That corrupted the Jacobian and collapsed the trust region to a spurious "converged" stop. Rejected probes are now retried in the opposite direction (the column is zeroed if both fail), and stops close to rejections are flagged. Infeasible starting points are recorded instead of crashing.
- **Projection check range.** A 0.8% fast-vs-direct projection difference occurred only at 80 pc, beyond the data (68 pc). Validation now covers the data radii.
- **Test isolation.** One test module set AGAMA's global units to kpc, so later DF tests refused to run (25 failures and errors in the full suite). A fixture now restores units; the full suite passes (630 passed, 5 skipped).
- **Remaining numerical limitation.** In strongly DM-dominated regions ($M_\star \sim 1.4 \times 10^6\,\mathrm{M_\odot}$, $\rho_{20} \sim 8$) the self-consistency iteration can diverge. The normalization tolerance also rejected some extreme starts. Neither affects the minima reported here.

Withdrawn: Gaia error floors

Two diagnostic fits added 0.01 and 0.02 mas yr^{-1} in quadrature to the Gaia errors. They showed that the size of the combined misfit depends strongly on the Gaia errors ($\chi^2_{\rm kin}/N$ falls from 5.12 to 2.09 with 0.02 mas yr^{-1}), while the anisotropy conflict persists. Such floors are ad hoc, not a calibrated error model. They have been withdrawn from all model assessment and will not be used. The error-model question is left to a separate, calibrated treatment.

8 Conclusions and next steps

1. The single-transition regularized DF cannot fit the combined ω Cen data. The inner and outer data demand radial and tangential anisotropy respectively. This is established by converged multi-start fits, subset fits and an independent Jeans analysis.
2. The two-transition DF, with no halo and no BH, fits the kinematics better than the best existing Jeans model ($\chi^2_{\rm kin} = 159.2$ vs 190.0 for 89 bins). Its anisotropy follows the Jeans turnover inside ~ 30 pc, and its DF is smooth. It is a structural near-pass, not a statistical pass.
3. The remaining misfits are localized at the MUSE outer edge and in the 4–20 pc tangential PMs, where the rotation correction is largest. The photometric misfit appears to be dominated by internal scatter in the compilation.
4. Adding a halo trades against the anisotropy. Its preference is not interpretable yet; no DM constraint is claimed.

Objective 1 (fit), proposed. (i) Model rotation with an odd-L_z DF component and fit the mean velocities jointly, instead of subtracting published curves. (ii) Calibrate the Gaia (and HST) error models from literature systematics or from the HST/Gaia overlap, as a separate data task whose method needs approval. (iii) Examine the photometric compilation for internally inconsistent sources. A central BH, separate tracer and mass DFs, and an axisymmetric DF remain later options, to be introduced only when the data demand them.

Objective 2 (DM), proposed. (i) A ρ_{20} profile likelihood with the two-transition DF. (ii) Realistic mocks (v3) from the best fits, with noise, to validate identifiability and coverage. (iii) Intervals only after the likelihood is calibrated.

A Reproducibility

Table 7: Batches reported here (under `results/df/`). Each batch freezes its source code, configuration, tests and inputs.

Batch	Commit	Content
`observed_df_20260923`	`bd17811`	One transition, default bounds, 2 branches × 2 starts
`observed_df_wide_20260923`	`5486309`	One transition, wide bounds, 2 branches × 6 starts
`dfdiag_{hst_muse,gaia,freedist}_20260923`	`1e246bd`	Diagnostic subset and distance fits
`dftwo_20260923`	`f6ce414`	Two transitions, no halo, 6 starts
`dftwo_halo_20260923`	`bb40db1`	Two transitions + free cored halo, 6 starts

The driver is `bin/run_compact_df_recovery.py` (`prepare-real`, `run`). The figures come from `bin/plot_observed_df_fits.py`, `plot_beta_profiles.py`, `plot_action_coverage.py`,

`plot_df_structure.py`, `plot_best_fit_performance.py` and `plot_central_bh_evidence.py`. Plotted data and checksums are in `results/plot_data/`. Markdown records are in `docs/OBSERVED_DF_FLEXIBILITY.md` and `docs/DF_ANISOTROPY_DIAGNOSTICS.md`. Example of the two-transition run:

```
PY=python
$PY bin/run_compact_df_recovery.py prepare-real --out results/df/NAME \
  --two-transition --branches no_halo --n-starts 4 --stop-delta 0.01 \
  --jacobian-seed base --probe-workers 5 --iteration-tolerance 5e-5 \
  --bound stellar.J0=30:800 --bound stellar.J_a=5:500 \
  --bound stellar.b_out=-2:2 --bound matter.r_s=5:500
$PY bin/run_compact_df_recovery.py run --out results/df/NAME \
  --workers 2 --max-calls 600 --detach
```

References

H. Baumgardt and E. Vasiliev. Accurate distances to Galactic globular clusters through a combination of Gaia EDR3, HST, and literature data. *MNRAS*, 505:5957–5977, 2021. doi: 10.1093/mnras/stab1474.

James Binney. Distribution functions for spheroids. *MNRAS*, 549(1):stag854, June 2026. doi: 10.1093/mnras/stag854.

M. Häberle, N. Neumayer, A. Seth, et al. Fast-moving stars around an intermediate-mass black hole in ω Centauri. *Nature*, 631:285–288, 2024. doi: 10.1038/s41586-024-07511-z.

Maximilian Häberle, N. Neumayer, C. Clontz, A. C. Seth, P. J. Smith, S. Kamann, R. Pechetti, M. S. Nitschai, M. Alfaro-Cuello, H. Baumgardt, A. Bellini, A. Feldmeier-Krause, N. Kacharov, M. Libralato, A. P. Milone, S. O. Souza, G. van de Ven, and Z. Wang. oMEGACat. VI. Analysis of the Overall Kinematics of Omega Centauri in 3D: Velocity Dispersion, Kinematic Distance, Anisotropy, and Energy Equipartition. *ApJ*, 983(2):95, April 2025. doi: 10.3847/1538-4357/adbe67.

L. Lindegren, S. A. Klioner, J. Hernández, et al. Gaia Early Data Release 3. The astrometric solution. *A&A*, 649:A2, 2021. doi: 10.1051/0004-6361/202039709.

Renuka Pechetti, Sebastian Kamann, Davor Krajnović, Anil Seth, Glenn van de Ven, Nadine Neumayer, Stefan Dreizler, Peter M. Weilbacher, Sven Martens, and Florence Wragg. ω Centauri: a MUSE discovery of a counter-rotating core. *MNRAS*, 528(3):4941–4957, March 2024. doi: 10.1093/mnras/stae294.

S. C. Trager, I. R. King, and S. Djorgovski. Catalogue of Galactic globular-cluster surface-brightness profiles. *AJ*, 109:218, 1995.

Eugene Vasiliev. AGAMA: action-based galaxy modelling architecture. *MNRAS*, 482(2):1525–1544, January 2019. doi: 10.1093/mnras/sty2672.

Eugene Vasiliev and Holger Baumgardt. Gaia EDR3 view on galactic globular clusters. *MNRAS*, 505(4):5978–6002, August 2021. doi: 10.1093/mnras/stab1475.